PT.SUMITOMO ELECTRIC WINTEC INDONESIA

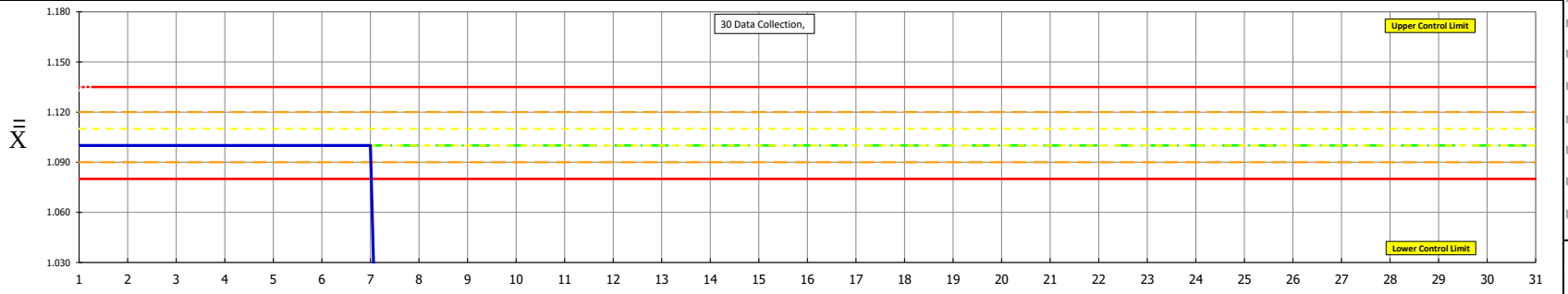
$\overline{X}$ - R CONTROL CHART

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

MACHINE	SECTION	OPERATION	DATE CONTROL LIMITS CALC.	SPECIFICATION	COURSE NO.	CONTROL ITEM (Δ) YES / NO
Softability	QC	Softability		min max 1.317		O
	MONTH	CHARACTERISTIC	SAMPLE SIZE / FREQUENCY	TYPE / SIZE : D2IDW4-B-2 / 0.70		
	May-25	Kelenturan	n = 3	PIC : YAYAT S.		

X = AVERAGE $\overline{X}$ = 1.100 U.C.L. = $\overline{\overline{X}} + A_2\overline{R}$ = 1.1350 L.C.L. = $\overline{\overline{X}} - A_2\overline{R}$ = 1.0800

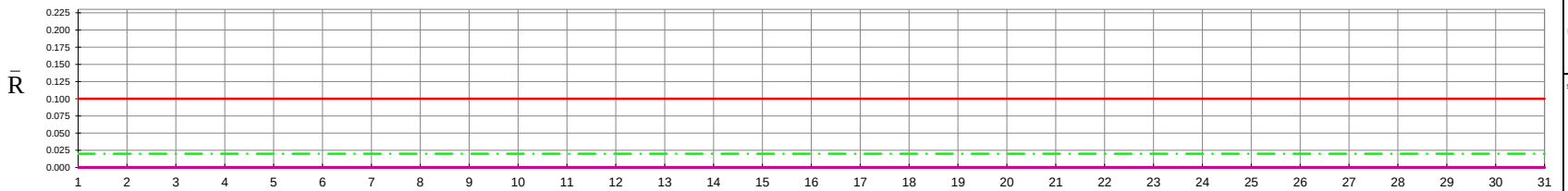
AVERAGE (X BAR CHART)



$\overline{R}$ = AVERAGE R = 0.020 U.C.L. = $D_4\overline{R}$ = 0.100 L.C.L. = $D_3\overline{R}$ = 0.00

RANGES (R - CHART)

Special Cause Marker Legend:
Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■



SPECIAL CAUSES

- ☑ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ☑ TEST 2: 7 POINTS IN A ROW ON SAME SIZE OF CENTER LINE
- ☑ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ☑ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ☑ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ☑ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

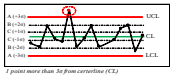
1. _____
2. _____
3. _____
4. _____
5. _____

1.100
1.100

SUBGROUP SIZE	A_2	d_2	D_3	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

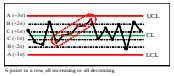
Special Cause

Titik berada di luar garis merah atau lebih dari 3 Ơ



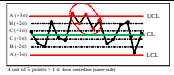
1 point more than 3σ from centerline (CL)

6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun



6 point in a row, all increasing or all decreasing

4 atau 5 titik lebih dari 1Ơ dengan posisi yang sama dan titik keluar dari garis merah atau 3Ơ



4 out of 5 points > 1 σ from centerline (same side)

Month																														
Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DATA 1	1.10	1.10	1.10	1.10	1.10	1.10	1.10																								
DATA 2	1.10	1.10	1.10	1.10	1.10	1.10	1.10																								
DATA 3	1.10	1.10	1.10	1.10	1.10	1.10	1.10																								
DATA 4																															
DATA 5																															
DATA 6																															
DATA 7																															
N	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, X	1.10	1.10	1.10	1.10	1.10	1.10	1.10	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range	0.000	0.000	0.000	0.000	0.000	0.000	0.000																								
Moving Range		0.002	0.768	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SHIFT 1	AHS	yat	yat		yat		R	R	yat	yat				R		yat	yat	yat	yat	yat			yat	yat	yat	yat					
SHIFT 2																															

CALCULATION

$\overline{X}$ = 1.1000
$\overline{R}$ = 0.0000
s_{ST} = 0.0000 s_{LT} = 0.000

THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED

C_{PK} = MIN [(X - LSL)/3s_{ST} , (USL - X)/3s_{ST}]
= MIN [#DIV/0! , #DIV/0!]
= **#DIV/0!**

Cp = [(USL - LSL)/6s_{ST}]
= [#DIV/0!]

CHECKED BY	APPROVED BY